

# CS5187: VISION AND IMAGE

---

## Effective Term

Semester A 2025/26

## Part I Course Overview

### Course Title

Vision and Image

### Subject Code

CS - Computer Science

### Course Number

5187

### Academic Unit

Computer Science (CS)

### College/School

College of Computing (CC)

### Course Duration

One Semester

### Credit Units

3

### Level

P5, P6 - Postgraduate Degree

### Medium of Instruction

English

### Medium of Assessment

English

### Prerequisites

CS3334 Data Structures or  
CS4335 Design and Analysis of Algorithms, or equivalent

### Precursors

Nil

### Equivalent Courses

Nil

### Exclusive Courses

Nil

## Part II Course Details

### Abstract

This course introduces algorithms in computer vision and image processing so as to develop students with basic knowledge to explain how computer could understand the visual world. The course describes visual understanding from the perspective of low-level image processing mid-level statistical inferencing and high-level vision recognition. The topics include feature extraction image segmentation, object recognition, motion analysis and scene understanding, along with real-world applications that vision algorithms have been successfully applied.

#### Course Intended Learning Outcomes (CILOs)

| CILOs |                                                                                                     | Weighting (if app.) | DEC-A1 | DEC-A2 | DEC-A3 |
|-------|-----------------------------------------------------------------------------------------------------|---------------------|--------|--------|--------|
| 1     | Perform low-level image processing for analysis of image properties.                                |                     |        | x      |        |
| 2     | Apply computer vision algorithms for scene/object understanding and 3D reconstruction.              |                     |        | x      |        |
| 3     | Assess and apply different computer vision and image processing approaches for real-world problems. |                     | x      |        |        |
| 4     | Design and implement computer vision and image processing algorithms for innovative applications.   |                     | x      | x      | x      |

#### A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

#### A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

#### A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

#### Learning and Teaching Activities (LTAs)

| LTAs |         | Brief Description                                                                                                                                                                                                                                      | CILO No. | Hours/week (if applicable) |
|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------|
| 1    | Lecture | Students will engage in lectures on the introduction of computer vision and image processing techniques, and related real-world applications such as feature matching, image enhancement, object recognition, motion analysis and scene understanding. | 1, 2, 3  | 2 hours/ week              |

|   |            |                                                                                                                                                                                                                                                                                                                                                                     |         |              |
|---|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------|
| 2 | Tutorial   | Students will work on a different problem set each week during the tutorial sessions, through which they can discover the main characteristics of different computer vision and image processing techniques and integrate them for real-world problems. They will also be invited to present their solutions, and the class will be encouraged to provide comments. | 1, 2, 3 | 1 hour/ week |
| 3 | Assignment | The students will implement selected computer vision and image processing approaches, apply these approaches to real-world problems, and interpret the results. In this way, students can analyse the performance of different approaches.                                                                                                                          | 1, 2, 3 |              |
| 4 | Project    | The students will create a new system design and implement appropriate computer vision and image processing approaches for innovative applications. The students will apply the principles they have learnt from the course for their design.                                                                                                                       | 4       |              |

**Assessment Tasks / Activities (ATs)**

| ATs |             | CILO No. | Weighting (%) | Remarks ("- " for nil entry)                                                           | Allow Use of GenAI? |
|-----|-------------|----------|---------------|----------------------------------------------------------------------------------------|---------------------|
| 1   | Assignments | 1, 2     | 30            | Expect to have two assignments with programming elements for algorithm implementation. | Yes                 |
| 2   | Project     | 4        | 20            | Can be in groups of 2–3 students per project                                           | Yes                 |

**Continuous Assessment (%)**

**Examination (%)**

50

**Examination Duration (Hours)**

2

**Minimum Examination Passing Requirement (%)**

30

**Additional Information for ATs**

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

**Assessment Rubrics (AR)**

**Assessment Task**

Assignment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

The ability to implement and assess the effectiveness of different algorithms and techniques.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not even reaching marginal level

---

**Assessment Task**

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

The ability and creativity in designing and implementing appropriate algorithms and techniques for innovative applications.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not even reaching marginal level

---

**Assessment Task**

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

The extent to which the students can understand the algorithms and techniques, apply them with appropriate modification or design new solutions for different applications, and evaluate their performances.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not even reaching marginal level

---

**Assessment Task**

Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

The ability to implement and assess the effectiveness of different algorithms and techniques.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B) Significant

**Marginal**

(B-, C+, C) Moderate to Basic

**Failure**

(F) Not even reaching marginal level

---

**Assessment Task**

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

The ability and creativity in designing and implementing appropriate algorithms and techniques for innovative applications.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B) Significant

**Marginal**

(B-, C+, C) Moderate to Basic

**Failure**

(F) Not even reaching marginal level

**Assessment Task**

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

The extent to which the students can understand the algorithms and techniques, apply them with appropriate modification or design new solutions for different applications, and evaluate their performances.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B) Significant

**Marginal**

(B-, C+, C) Moderate to Basic

**Failure**

(F) Not even reaching marginal level

## Part III Other Information

**Keyword Syllabus**

Feature extraction and alignment; image segmentation; image enhancement / editing; 3D reconstruction; face detection and recognition; object recognition; motion analysis; scene understanding.

**Reading List****Compulsory Readings**

| Title |                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------|
| 1     | Richard Szeliski, Computer Vision: Algorithms and Applications Springer; 2011.                            |
| 2     | D. Forsyth and J. Ponce, Computer Vision: A Modern Approach 2nd Ed, Prentice Hall (2011)                  |
| 3     | Simon J. D. Prince, Computer Vision: Models, Learning, and Interference Cambridge University Press, 2012. |
| 4     | R. Gonzalez and R. Woods, Digital Image Processing 3rd Ed, Prentice Hall (2007)                           |

**Additional Readings**

| Title |     |
|-------|-----|
| 1     | Nil |